Firstly, it leads to the removal of colour artifacts popularly called "coloured jaggies" associated with this process. Also, since very little aliasing occurs, there's no need for anti-aliasing filters, as are necessary in Bayer sensors. With the help of micro lenses, photodiodes for a particular colour integrate the optical image over a region the size of spacing of sensors for that colour. Though this removes the colour artifacts, colour separation by silicon penetration depth leads to increased cross contamination in between colour layers, in turn compromising colour accuracy.

Another advantage with the Fovean X3 is that the camera manages to detect a higher portion of photons entering it than would be possible with a mosaic sensor, in which two colours are absorbed by each of the filters overlaying a photo-site so that a single primary colour passes through. This causes a significant reduction in the total amount of light gathered, destroying critical information regarding the colour of the light that strikes on each sensor element. Though there's a higher light gathering ability in Fovean X3, the different individual colours don't receive a sharp response from the separate layers raising the need for removal of common mode signals through "matrixing" of the colour indicating information in the sensors' raw data. Though this produces the colour data within appropriate colour spaces, it also enhances noise in low-light scenarios.

Size

Most SLRs have sensors the size of a frame of APS-C film, with crop factors around 1.5-1.6. Olympus and Panasonic though are notable exceptions to the popular four-thirds system of cameras, as they employ smaller sensors with typical crop factor values around 2.0. Full-frame sensors around the size of a 35mm film frame are also used by a few professional dSLR cameras. Now you may have noticed that markets are full of marketing terms describing sensor formats such as "full frame" digital SLR format, which means that the dimension for this particular sensor nearly equals that of 35mm films (that is 36 mm × 24 mm). Or say Canon's APS-H format, which indicates high-speed pro-level dSLR with crop factors 1.33. APS-C refers to a similarly sized range including Nikon DX format, Pentax, Fuji etc among others with crop factor 1.5 and even entry-level dSLR formats from Canon with crop 1.6. As compared to an APS-C sensor, a full-framed sensors' production costs may be upto twenty times higher. On a generic 8-inch silicon wafer used for manufacturing, 112 APS-C sensors can fit in, while only around 30 full-frame sensors can be accommodated on the same sheet. Also the